Data Warehousing/ Visualization Group Assignment

Master of Data Analytics

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1. Method and Data Preparation

This project focuses on designing an interactive business intelligence dashboard using Power BI to analyze sales performance across products, customers, and regions. The dataset provided consisted of five CSV files: FactOrders, DimProduct, DimCustomer, DimRegion, and DimDate. These files were integrated into a star schema data model to support efficient analytical queries andvisualization.

The FactOrders table contained transactional sales data including order details, sales amount, quantity, profit, discount, shipping mode, and order dates. The dimension tables provided descriptive attributes such as product category and sub-category (DimProduct), customer demographics and geographic information (DimCustomer), region data (DimRegion), and a structured calendar hierarchy (DimDate).

Data preparation involved importing the datasets into Power BI and performing basic ETL processes. These steps included validating column data types, checking for missing values, and ensuring that keys such as ProductID, CustomerID, and Region were consistent across tables. The DimDate table was marked as the official date table to enable time intelligence features and consistent filtering throughout the dashboard.

A star schema was implemented where FactOrders served as the central fact table and the four dimension tables provided descriptive attributes. Relationships were created between FactOrders and each dimension table to support slicing and aggregation of measures such as revenue, profit, and units sold.

2. Data Model

The data model follows a standard star schema design which is widely used in data warehousing and business intelligence systems. FactOrders acts as the central table containing quantitative metrics, while the dimension tables provide contextual information.

Key relationships include:

- FactOrders[ProductID] → DimProduct[ProductID]

- FactOrders[CustomerID] → DimCustomer[CustomerID]

- FactOrders[Region] → DimRegion[Region]

- FactOrders[OrderDate] → DimDate[Date]

This model supports flexible filtering and enables cross-analysis across products, customers, regions, and time periods.

Several DAX measures were created to support analysis, including Total Revenue, Total Units, Average Selling Price (ASP), Total Profit, and Profit Margin. These measures allow stakeholders to analyze financial performance from multiple perspectives.

Key DAX Measures are as follows:

- Total Revenue = SUM(FactOrders[Sales])
- Total Units = SUM(FactOrders[Quantity])
- Average Selling Price (ASP) =DIVIDE([Total Revenue], [Total Units])
- Total Profit = SUM(FactOrders[Profit])
- Profit Margin =DIVIDE([Total Profit], [Total Revenue])

3. Dashboard Design and Key Visuals

The dashboard was designed as a multi-page report to support different analytical perspectives. The first page, Executive Overview, presents high-level KPIs including Total Revenue, Total Units, ASP, and Profit Margin. A revenue trend chart and map visualization show geographic distribution of sales by country. The map visual highlights geographic sales concentration and helps identify countries generating the highest revenue.

The Customer Analysis page focuses on customer behavior. Key visuals include a bar chart of the Top 10 customers by revenue, a revenue-by-state chart, and a weekly revenue trend. These visuals help identify high-value customers and regional demand patterns.

The Product Performance page analyzes product categories and individual product performance. A matrix visual shows the Top 10 products by revenue along with supporting metrics such as units sold, ASP, and profit margin. This page also includes a scatter plot comparing ASP and product rating (derived from profit margin) with bubble size representing units sold. Where X-axis: ASP, Y-axis: Rating and Bubble: Units sold. This visualization highlights the relationship between pricing strategy and product performance.

A drill-through page was implemented allowing users to right-click a product or customer and navigate to a detailed analysis page showing revenue, units, ASP, shipping mode distribution, and time trends for the selected item.

Interactive features were incorporated to improve usability. Year and Quarter slicers were synchronized across the Executive Overview, Customer Analysis, and Product Performance pages using Power BI’s Sync Slicers functionality. This allows users to apply time-based filters consistently across all views.

Bookmarks were also implemented to support two user personas: Retail Buyer and Product Manager. The Retail Buyer view highlights customer and shipping insights, while the Product Manager view emphasizes product performance metrics such as ASP and profit margin. Where Retail Buyer focuses on: customer demand, shipping mode, regional sales and Product Manager focuses on product category performance, ASP, profit margin.

Key visuals are:

Figure 01: Executive Overview Dashboard

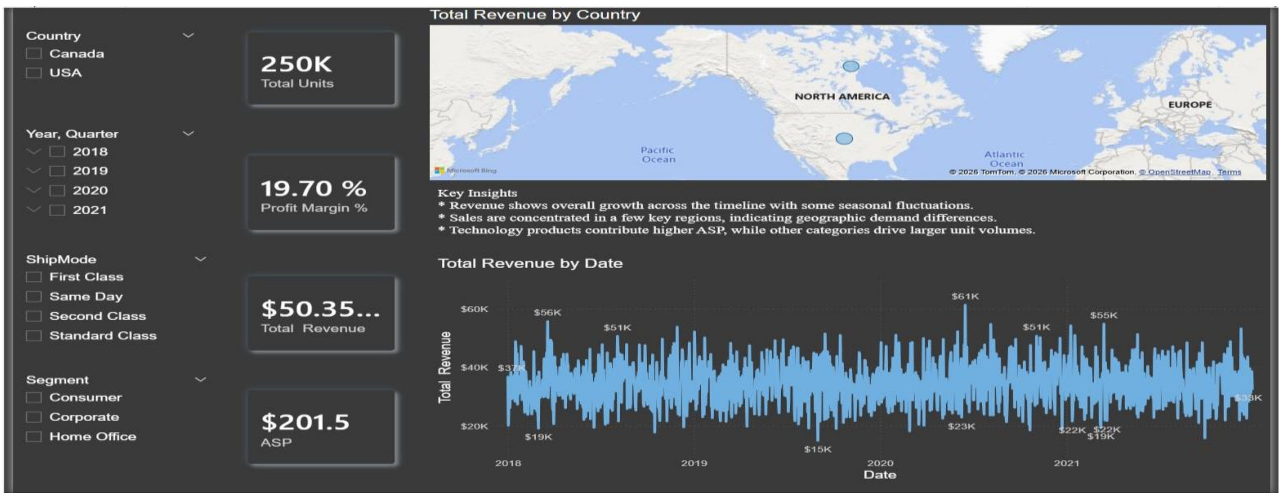


Figure 02: Customer Analysis Page

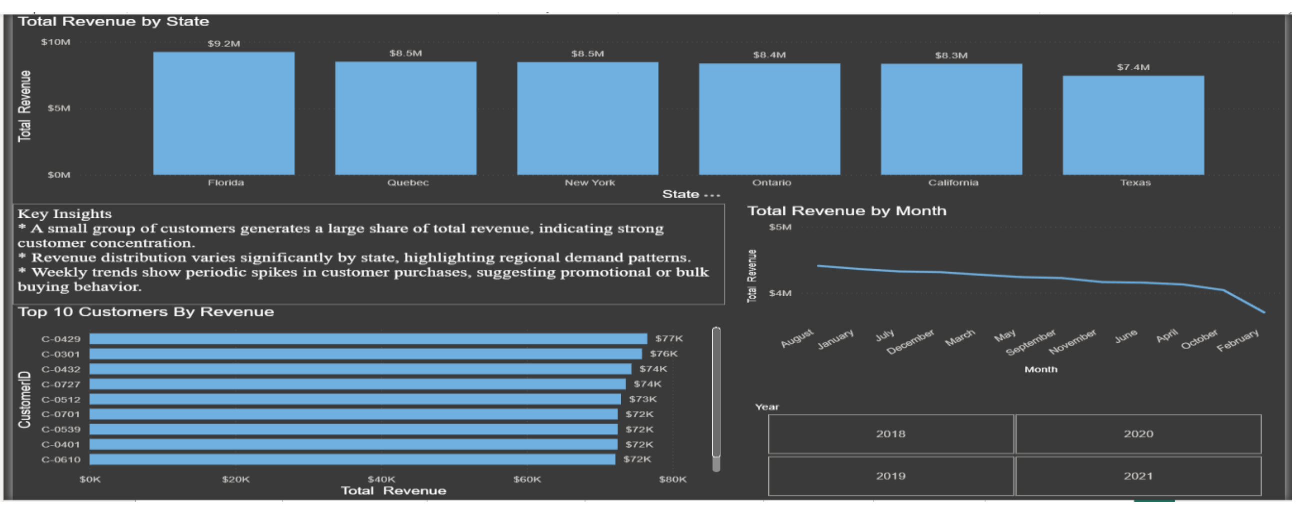


Figure 03: Product Performance Page

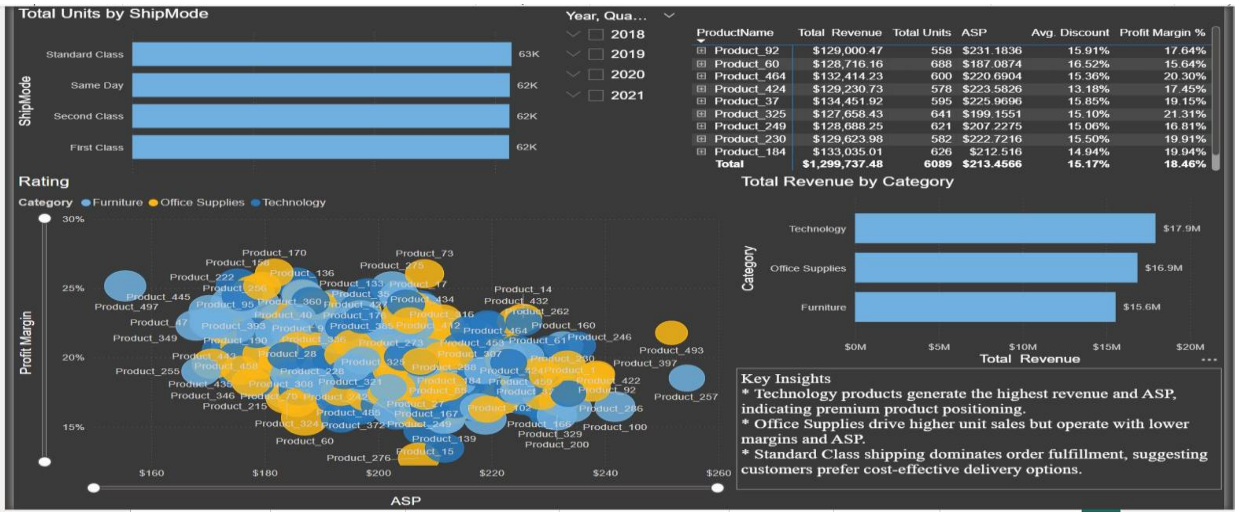
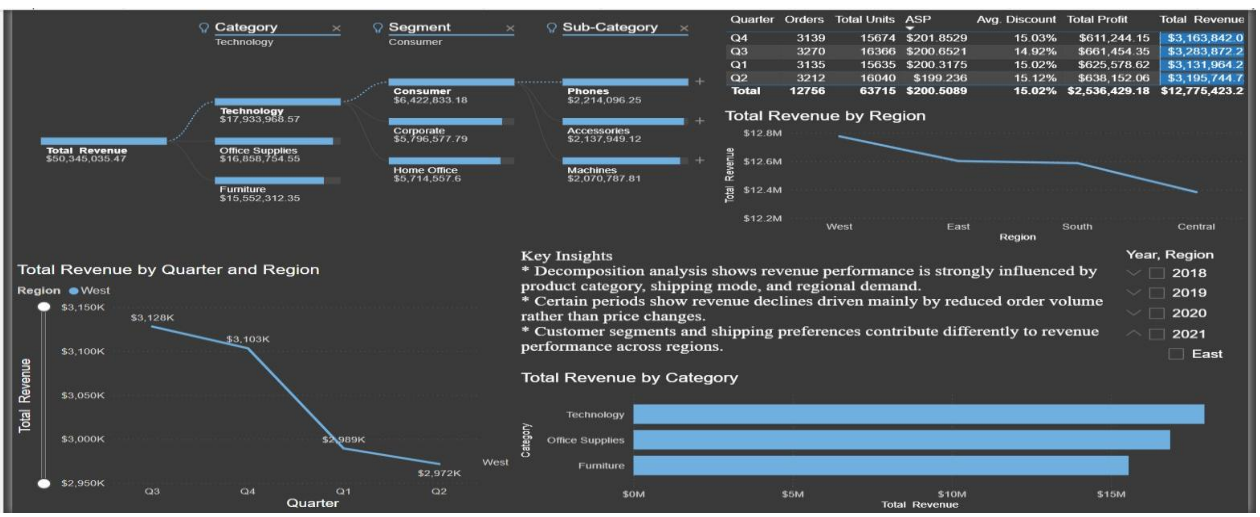


Figure 04: Diagnostic Analysis Page



4. Diagnostic Analysis

A dedicated diagnostic page was created to investigate anomalies and identify potential drivers of performance changes. A decomposition tree visual was used to analyze Total Revenue across multiple dimensions including category, sub-category, ship mode, state, and customer segment.

This visual enables interactive root-cause analysis by progressively breaking down revenue contributions.

For example, Firstly, a decline in revenue during certain quarters can be examined by drilling down into product categories and shipping modes. Supporting visuals such as revenue trends and ship mode distribution charts help provide additional context. A table comparing metrics such as Orders, Units, ASP, Discount, and Profit across quarters was also included to identify operational drivers of revenue changes. Secondly, Quarterly revenue dip in Q3 for Technology products which may Possible cause the increased discounts or shipping delays. Further, High revenue but low profit in Furniture category which may Possible cause the higher discounts and shipping costs.

5. Security and Interactivity

To demonstrate secure data access, Row-Level Security (RLS) was implemented in the dashboard. A role named 'RegionManager' was created using Power BI’s Manage Roles functionality. A filter was applied on the DimRegion table to restrict data visibility to a specific region (e.g., West). This ensures that users assigned to the role only see data relevant to their designated region.

The role was tested using Power BI’s 'View As' feature to confirm that only the filtered regional data was displayed. This example demonstrates how organizations can enforce secure and role-based access to analytical dashboards.

6. Recommendations

Based on the analysis, several actionable insights were identified. First, a small number of customers contribute a significant portion of revenue, suggesting that customer retention strategies for high-value clients should be prioritized. Second, product performance varies across categories, with certain categories achieving higher ASP and profit margins. Pricing and promotional strategies should be adjusted accordingly.

Third, shipping mode analysis revealed that standard shipping dominates order fulfillment. Organizations should evaluate whether alternative shipping options could improve customer satisfaction while maintaining operational efficiency.

Overall, the dashboard provides a comprehensive analytical view of business performance. The combination of interactive visuals, synchronized filters, diagnostic tools, and secure access controls supports informed decision-making for multiple stakeholders.

7. Conclusion

This project demonstrates how Power BI can be used to transform raw transactional data into meaningful business insights through effective data modeling and visualization. By integrating multiple datasets into a star schema structure, the analysis enabled efficient exploration of sales performance across products, customers, regions, and time periods. The interactive dashboard provided key performance indicators such as total revenue, units sold, average selling price, and profit margin, allowing stakeholders to quickly evaluate overall performance trends.

The use of advanced visuals, synchronized slicers, and diagnostic tools supported deeper analysis of customer behavior, product performance, and operational factors affecting revenue. In addition, the implementation of row-level security ensured controlled access to sensitive data for different user roles. Overall, the dashboard provides a scalable and interactive analytical solution that supports informed decision-making and strategic business planning.